

BRAINWARE UNIVERSITY
SCHOOL OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
Bachelor of Technology in Computer Science & Engineering – 2025
SEMESTER–III

Course Code: BTS30102

Course Name: Database Management Systems

Weekly Contact Hours: 3L + 1T

Credit(s): 4

Total Allotted Hours: 60H (45L + 15T)

Course Objective:

In the modern age data is most valuable part. To extract some information from a huge amount of data and store for near future to collect some knowledge also an important area of development. In this scenario data management plays a vital role. From this course learners can understand every point of data managements and also it can implement its day to day life.

Pre-requisite(s): Set theory concept of mathematics.

Course Outcome: After the completion of the course, students would be able to:

CO1: Define the various aspects of data definition, classify different data models and cite the example of entity relationship diagram.

CO2: Describe and compare the several relational query languages, query processing and query optimization using open source and commercial DBMS.

CO3: Define different storage structure like B-Tree, B+ Tree to analyze the storage policy.

CO4: Explain and Analyze the transaction process to interpret serializability of scheduling process and concurrency control policy.

CO5: Evaluate and formulate the security policy of a database in conjunction with the advancement of database design.